

Data types and type casting

1. Data Types in JavaScript

Detailed Explanation:

- **Data Types:** JavaScript is a loosely typed language, meaning variables can store different types of values over time. Data types in JavaScript are divided into:
 - **Primitive types:** Boolean, Number, BigInt, String, Symbol, undefined, and null.
 - **Complex types:** Objects and Arrays.
- **Primitive Types:** These types represent a single value (atomic).
 - **Boolean:** Represents logical values (true or false).
 - **Number:** Represents both integers and floating-point numbers.
 - **BigInt:** Used for representing integers larger than the range supported by Number.
 - **String:** Represents textual data.
 - **undefined:** The default value of uninitialized variables.
 - **Symbol:** Represents a unique identifier.
 - **null:** Represents the intentional absence of any object value.

Code Snippet:

javascript

Copy code

```
let isJavaScriptFun = true; // Boolean
```

```
let age = 30; // Number
```

```
let largeNumber = 12345678901234567890n; // BigInt
```

```
let name = "Stavan"; // String
```

```
let notDefined; // undefined
```

```
let uniqueKey = Symbol('id'); // Symbol
```

```
let emptyValue = null; // null
```

```
console.log(typeof isJavaScriptFun); // Output: boolean
```

```
console.log(typeof age); // Output: number
```

```
console.log(typeof largeNumber); // Output: bigint
```

```
console.log(typeof name); // Output: string
```

```
console.log(typeof notDefined); // Output: undefined
```

```
console.log(typeof uniqueKey); // Output: symbol
```

```
console.log(typeof emptyValue); // Output: object (known issue with JavaScript)
```

In-depth Questions:

1. **What are the six primitive data types in JavaScript?**

- **Answer:** Boolean, Number, BigInt, String, Symbol, undefined, and null.

2. **What is the purpose of BigInt in JavaScript?**

- **Answer:** BigInt allows for representing integers that are too large to be handled by the Number type.

Practice Question:

- **True or False:** null is a primitive data type but returns "object" when checked using typeof.
 - **Answer:** True.
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2. The typeof Operator

Detailed Explanation:

- The **typeof** operator is used to determine the data type of a variable or literal.
 - For example, `typeof 42` returns "number".
- **Common typeof returns:**
 - "undefined" for uninitialized variables.
 - "object" for null.
 - "number", "string", "boolean", "bigint", "symbol" for respective primitive types.
 - "function" for functions.

Code Snippet:

```
javascript
```

Copy code

```
let age = 25;
```

```
console.log(typeof age); // Output: number
```

```
let name = "Alice";  
console.log(typeof name); // Output: string
```

```
let notDefined;  
console.log(typeof notDefined); // Output: undefined
```

```
let isHappy = true;  
console.log(typeof isHappy); // Output: boolean
```

In-depth Questions:

1. **What does `typeof null` return, and why?**
 - **Answer:** It returns "object", which is a known quirk in JavaScript due to legacy reasons.
2. **What will `typeof functionName` return in JavaScript?**
 - **Answer:** It will return "function".

Practice Question:

- **What will `typeof 123n` return?**
 - **Answer:** "bigint".
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3. Primitive Data Types

3.1 Boolean

- **Boolean** is used to represent logical values (true or false). It is commonly used for control flow and decision-making.

javascript

Copy code

```
let isValid = true;
```

```
let isGameOver = false;
```

3.2 Number

- **Number** can represent both integer and floating-point numbers. Special values include Infinity, -Infinity, and NaN.

javascript

Copy code

```
let num = 100;  
let fraction = 3.14;  
let infiniteValue = 1 / 0; // Infinity
```

3.3 BigInt

- **BigInt** is used for representing very large integers. You can declare it with an n at the end of a number.

javascript

Copy code

```
let bigNumber = 12345678901234567890n;
```

3.4 String

- **String** is used to represent text and can be created using single (') or double (") quotes.

javascript

Copy code

```
let firstName = "John";  
let lastName = 'Doe';
```

3.5 Undefined and Null

- **undefined** is the default value for variables that are declared but not initialized.

javascript

Copy code

```
let undefinedVar;  
console.log(undefinedVar); // undefined
```

- **null** is explicitly assigned to a variable to indicate "no value."

javascript

Copy code

```
let emptyVar = null;
```

4. Type Casting and Conversion

Detailed Explanation:

- **Type Casting:** JavaScript allows converting values from one type to another. This can be done explicitly using constructor functions like `String()`, `Number()`, and `Boolean()`, or implicitly.

Code Snippet:

javascript

Copy code

```
// Explicit conversion  
let num = 100;  
  
let strNum = String(num); // Convert number to string  
  
console.log(typeof strNum); // Output: string
```

```
let boolValue = Boolean(0); // Convert 0 to Boolean  
  
console.log(boolValue); // Output: false
```

In-depth Questions:

1. **What is type casting in JavaScript?**
 - **Answer:** Type casting refers to converting a value from one data type to another, such as converting a number to a string or a boolean to a number.
2. **What does `Boolean(0)` return?**
 - **Answer:** `false` (since 0 is falsy).

5. Implicit and Explicit Type Conversions

Detailed Explanation:

- **Implicit Conversion:** JavaScript automatically converts types in certain situations, such as when adding a number and a string.

javascript

Copy code

```
let result = 42 + "1"; // Implicit conversion to string  
  
console.log(result); // Output: "421"
```

- **Explicit Conversion:** You can explicitly convert values using constructors like `Number()`, `String()`, and `Boolean()`.

javascript

Copy code

```
let num = Number("123");  
console.log(num); // Output: 123
```

Code Snippet:

javascript

Copy code

```
let str1 = 42 + "1"; // Implicit: number to string  
console.log(str1); // Output: "421"
```

```
let num1 = "42";
```

```
let convertedNum = Number(num1); // Explicit: string to number  
console.log(convertedNum); // Output: 42
```

In-depth Questions:

1. **What happens when you add a string to a number in JavaScript?**
 - **Answer:** JavaScript converts the number to a string and performs string concatenation.
 2. **What does `Number("text")` return?**
 - **Answer:** NaN (since "text" cannot be converted to a number).
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6. Conversion to String, Number, and Boolean

Conversion to String

javascript

Copy code

```
let num = 42;  
let str = String(num);  
console.log(str); // Output: "42"
```

Conversion to Number

javascript

Copy code

```
let str = "123";
```

```
let num = Number(str);  
console.log(num); // Output: 123
```

Conversion to Boolean

javascript

Copy code

```
let falsyValue = "";  
let booleanValue = Boolean(falsyValue);  
console.log(booleanValue); // Output: false
```

7. Tasks from the File

- **Task 1:** Create variables for Boolean, Number, BigInt, String, and undefined, using literals and constructor functions. Display the values and their types using string interpolation.

javascript

Copy code

```
let boolVal = true;  
let numVal = 42;  
let bigIntVal = 1234567890123456789n;  
let strVal = "Hello";  
let undefinedVal;  
  
console.log(`${boolVal} [${typeof boolVal}]`);  
console.log(`${numVal} [${typeof numVal}]`);  
console.log(`${bigIntVal} [${typeof bigIntVal}]`);  
console.log(`${strVal} [${typeof strVal}]`);  
console.log(`${undefinedVal} [${typeof undefinedVal}]`);
```